HEIGHTS ON PROJECTIVE SPACE, WITH SOLUTIONS

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To apply the descent theorem to $E(K)$, one needs a height on $E(K)$ with the Northcott property. Silverman builds it from a height on projective space [Sil09, §VIII.5]: a point of $\mathbb{P}^N(\overline{\mathbb{Q}})$ is large when its coordinates are large at the archimedean places or have large denominators at the non-archimedean ones. This worksheet works out the first properties of this height. Problem 1.5, the heart of the worksheet, proves Kronecker's theorem: the algebraic numbers of height 1 are 0 and the roots of unity. Problem 2.1 proves the inequalities for products and sums, and the behavior of the height under the Segre and d-uple embeddings. Problem 3.2 computes the height through the ideal generated by the coordinates, and shows on $\mathbb{Q}(\sqrt{-5})$ why the recipe of coprime integer coordinates does not extend to every number field. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and its subring $\overline{\mathbb{Z}}$ of algebraic integers.
- A number field K, its ring of integers $\mathcal{O}_K$, its set of places $\mathcal{P}_K$, and the subsets $\mathcal{P}_K^\infty$ and $\mathcal{P}_K^0$ of archimedean and non-archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$ and $v \in \mathcal{P}_K$, the number $|P|_v \coloneqq \max_i |x_i|_v$. It depends on the choice of coordinates.

How to use this. Allow about 25 minutes for Problem 1.5, 15 for Problem 2.1 and 20 for Problem 3.2. Problem 1.5 is the one not to skip. The `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

1. Kronecker's theorem

This section is [Sil09, Exercise 8.6], with the proof of [HS00, Corollary B.2.3.1]. The definitions and the first properties of the height are [Sil09, §VIII.5]. The last two items compare Kronecker's theorem with the enumeration of polynomials that proves it in small degree, and with algebraic numbers of small height.

Proposition 1.1 (Extension and product formulas [Sil09, (VIII.5.2), (VIII.5.3)])**.** *(1) Let L be a finite extension of K, and let $v \in \mathcal{P}_K$. Then $\sum_{w|v} n_w = [L : K]\, n_v$, the sum over the places $w \in \mathcal{P}_L$ that restrict to v.*
(2) Let $x \in K^\times$. Then $\prod_{v \in \mathcal{P}_K} |x|_v^{n_v} = 1$.

Definition 1.2 (Heights on projective space [Sil09, §§VIII.5, VIII.6])**.** The height relative to K of a point $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$ is

$$H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} |P|_v^{n_v}.$$

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Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and let K be a number field with $P \in \mathbb{P}^N(K)$. The height of P is $H(P) := H_K(P)^{1/[K:\mathbb{Q}]}$, and its logarithmic height is $h(P) := \log H(P)$. For $x \in \overline{\mathbb{Q}}$, let $H(x) := H([x:1])$ and $h(x) := \log H(x)$.

Proposition 1.3 (Properties of the height [Sil09, Proposition VIII.5.4]). *Let $P \in \mathbb{P}^N(K)$.*

(1) The height $H_K(P)$ does not depend on the choice of coordinates of P.
(2) The height satisfies $H_K(P) \geqslant 1$.
(3) Let L be a finite extension of K. Then $H_L(P) = H_K(P)^{[L:K]}$.

In particular, $H(P)$ does not depend on the choice of K, and $H(P) \geqslant 1$.

Theorem 1.4 (Northcott's theorem [Sil09, Theorem VIII.5.11]). *Let C and d be real numbers. Then the set of points $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(P) \leqslant C$ and $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant d$ is finite, where $\mathbb{Q}(P)$ is the field generated by the ratios x_i/x_j with $x_j \neq 0$ of the coordinates of P. In particular, the set $\{P \in \mathbb{P}^N(K) : H_K(P) \leqslant C\}$ is finite.*

Problem 1.5 (Kronecker's theorem).

This problem proves Kronecker's theorem: a nonzero algebraic number $x \in \overline{\mathbb{Q}}^\times$ has $H(x) = 1$ if and only if x is a root of unity. By (2), it is equivalent to Kronecker's original form: a nonzero algebraic integer $x \in \overline{\mathbb{Z}}$ with $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon \mathbb{Q}(x) \to \mathbb{C}$ is a root of unity.

(1) Show that every root of unity has height 1. Show that $H(0) = 1$, so the hypothesis $x \neq 0$ cannot be dropped.

Solution. **If $\zeta^n = 1$, then $|\zeta|_v^n = 1$ for every place v.** Let ζ be a root of unity, with $\zeta^n = 1$ for some $n \geqslant 1$, and let $K := \mathbb{Q}(\zeta)$. For every $v \in \mathcal{P}_K$, $|\zeta|_v^n = 1$ and $|\zeta|_v > 0$, so $|\zeta|_v = 1$. So every factor $\max(|\zeta|_v, 1)^{n_v}$ of $H_K([\zeta:1])$ is 1. Likewise every factor of $H_{\mathbb{Q}}([0:1])$ is $\max(0,1) = 1$. But 0 is not a root of unity.

(2) Let $x \in \overline{\mathbb{Q}}$ and $K := \mathbb{Q}(x)$. Show that $H(x) = 1$ if and only if $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K$, and that this holds if and only if $x \in \overline{\mathbb{Z}}$ and $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon K \to \mathbb{C}$. Show that $x = (3+4i)/5$ satisfies $|\tau(x)| = 1$ for both embeddings of $\mathbb{Q}(i)$, and compute $H(x)$.

Solution. **$H(x) = \sqrt{5}$.** Every factor $\max(|x|_v, 1)^{n_v}$ of $H_K([x:1])$ is at least 1, so the product is 1 if and only if every factor is 1, that is, $|x|_v \leqslant 1$ for every v. The elements of K with $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K^0$ form $\mathcal{O}_K$, and the archimedean absolute values are $|x|_v = |\tau(x)|$ for the embeddings $\tau\colon K \to \mathbb{C}$, each place coming from one embedding or from a pair of complex conjugate ones [HS00, Proposition B.1.3]. This gives the second equivalence.
For $x = (3+4i)/5$, the two embeddings send x to $(3 \pm 4i)/5$, of absolute value $\sqrt{9+16}/5 = 1$. In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$ and $3+4i = (2+i)^2$, so

$$x = \frac{(2+i)^2}{(2+i)(2-i)} = \frac{2+i}{2-i}.$$

So $x \notin \overline{\mathbb{Z}}$. At the place v of the prime $(2-i)$, of residue field $\mathbb{F}_5$, we have $n_v = 1$ and $|x|_v = 5$. At every other non-archimedean place, $|x|_v \leqslant 1$, and at the complex place $|x|_v = 1$. So $H_K([x:1]) = 5$ and $H(x) = 5^{1/2}$.

Remark. So the integrality in Kronecker's original statement (an algebraic integer all of whose conjugates lie on the unit circle is a root of unity) cannot be dropped. Problem 3.2 computes such heights in one step.

(3) Let $x \in \overline{\mathbb{Q}}$. Show that $H(x^n) = H(x)^n$ for every integer $n \geqslant 1$, and that $H(1/x) = H(x)$ if $x \neq 0$. Prove Kronecker's theorem.

SOLUTION. **The powers x^n lie in $\mathbb{Q}(x)$ and have height 1, so $x^r = x^s$ for some $r < s$.** Let $K := \mathbb{Q}(x)$. For every $v \in \mathcal{P}_K$, $\max(|x^n|_v, 1) = \max(|x|_v, 1)^n$, so $H_K([x^n : 1]) = H_K([x : 1])^n$. If $x \neq 0$, then $[1/x : 1] = [1 : x]$, and $|[1 : x]|_v = |[x : 1]|_v$, so $H(1/x) = H(x)$.

One direction of Kronecker's theorem is (1). Conversely, let $x \in \overline{\mathbb{Q}}^\times$ with $H(x) = 1$. Then $H_K([x^n : 1]) = 1$ for every $n \geqslant 1$, and the points $[x^n : 1]$ lie in $\mathbb{P}^1(K)$. By Theorem 1.4, they are finitely many, so $x^r = x^s$ for some $1 \leqslant r < s$. Since $x \neq 0$, $x^{s-r} = 1$.

(4) Let $f = t^d + a_1 t^{d-1} + \cdots + a_d \in \mathbb{Z}[t]$ be monic of degree $d \geqslant 1$, with all complex roots in the closed unit disc. Show that $|a_i| \leqslant \binom{d}{i}$ for every i. So there are finitely many such f of each degree. `Magma` lists those of degree 4 with $a_4 \neq 0$, and checks that each is a product of cyclotomic polynomials:

```
R<t> := PolynomialRing(Integers());
d := 4;
B := [[-Binomial(d, i)..Binomial(d, i)] : i in [1..d]];
S := [];
for c in CartesianProduct(B) do
  f := t^d + &+[c[i]*t^(d - i) : i in [1..d]];
  if c[d] ne 0 and forall{r : r in Roots(f, ComplexField(30)) |
                          Abs(r[1]) le 1 + 10^-10} then
    Append(~S, f);
  end if;
end for;
#CartesianProduct(B), #S;                          // 3159 24
forall{f : f in S | forall{g : g in Factorization(f) |
                           IsCyclotomicPolynomial(g[1])}};       // true
```

Use this output to prove Kronecker's theorem for every $x \in \overline{\mathbb{Q}}^\times$ with $[\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$, without Theorem 1.4.

SOLUTION. **The minimal polynomial g of x, times $(t-1)^{4-\deg g}$, is in the list.** Let $\alpha_1, \ldots, \alpha_d$ be the roots of f, with multiplicity. Then $a_i = (-1)^i e_i(\alpha_1, \ldots, \alpha_d)$, a sum of $\binom{d}{i}$ products of roots, each of absolute value at most 1. So $|a_i| \leqslant \binom{d}{i}$.

Let $x \in \overline{\mathbb{Q}}^\times$ with $H(x) = 1$ and $e := [\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$. By (2), $x \in \overline{\mathbb{Z}}$, and its conjugates lie in the closed unit disc. So its minimal polynomial $g \in \mathbb{Z}[t]$ is monic of degree e, with all roots in the disc, and $g(0) \neq 0$ since $x \neq 0$. Then $f := g \cdot (t-1)^{4-e}$ is monic of degree 4, with all roots in the disc and $f(0) \neq 0$. So f is one of the 24 polynomials of the list, and its irreducible factor g is cyclotomic. So x is a root of unity.

Remark. This is Kronecker's own proof: the minimal polynomials of the powers x^n of such an x all satisfy the same bound on their coefficients, so they are finitely many. The bound $|a_i| \leqslant \binom{d}{i}$ plays the role of [Sil09, Theorem VIII.5.9] in the proof of Theorem 1.4. The enumeration is exponential in d: for $d = 6$ it already takes minutes.

(5) Let $d \geqslant 1$, and let $2^{1/d}$ be the real root of $t^d - 2$. Show that $h(2^{1/d}) = (\log 2)/d$. So h takes arbitrarily small positive values on $\overline{\mathbb{Q}}^\times$. `Magma` checks this, and computes the logarithmic height of a root b of Lehmer's polynomial:

```
R<t> := PolynomialRing(Integers());
[RealField(10) | d*AbsoluteLogarithmicHeight(NumberField(t^d - 2).1)
                 : d in [2..5]];       // [ 0.6931471805, 0.6931471805, ... ]
```

```
RealField(10)!Log(2);                    // 0.6931471805
f := t^10 + t^9 - t^7 - t^6 - t^5 - t^4 - t^3 + t + 1;
IsIrreducible(f), IsCyclotomicPolynomial(f);          // true false
K<b> := NumberField(f);
RealField(10)!AbsoluteLogarithmicHeight(b);           // 0.01623576120
```

Explain why $h(b) > 0$, and compute $[\mathbb{Q}(b) : \mathbb{Q}]\, h(b)$.

Solution. **$h(2^{1/d}) = (\log 2)/d$, and $10\,h(b) \approx 0.1624$.** Let $\alpha := 2^{1/d}$ and $K := \mathbb{Q}(\alpha)$. The polynomial $t^d - 2$ is Eisenstein at 2, so $[K : \mathbb{Q}] = d$. Since $\alpha \in \mathcal{O}_K$, the non-archimedean factors of $H_K([\alpha : 1])$ are 1. The archimedean factors are $\max(|\tau(\alpha)|, 1)$, one for each of the d embeddings τ, a complex place contributing its two embeddings through $n_v = 2$. Each $\tau(\alpha)$ is $2^{1/d}$ times a root of unity, so $|\tau(\alpha)| = 2^{1/d} > 1$. So $H_K([\alpha : 1]) = (2^{1/d})^d = 2$, and $h(\alpha) = (\log 2)/d$.
The polynomial f is irreducible and not cyclotomic, so its root b is not a root of unity. By Kronecker's theorem, $h(b) > 0$. Here $[\mathbb{Q}(b) : \mathbb{Q}] = 10$, and $10\,h(b) \approx 0.1624$.

Remark. So there is no positive lower bound for h on the algebraic numbers that are not roots of unity. Lehmer asked whether there is one for $[\mathbb{Q}(x) : \mathbb{Q}]\, h(x)$, a question still open. The value $10\,h(b) = \log 1.17628\ldots$ of his polynomial is the smallest positive one known.

2. Basic properties of the height

This section is [Sil09, Exercise 8.8]. The last two items are [HS00, Proposition B.2.4(b), (c)], for the Segre embedding [Har77, Exercise I.2.14] and the d-uple embedding [Har77, Exercise I.2.12]. The heights are those of Definition 1.2.

Problem 2.1 (Basic properties of the height)**.**
In (1) and (2), let $N \geqslant 1$ and $x_1, \ldots, x_N \in \overline{\mathbb{Q}}$.

(1) Show that $H(x_1 x_2 \cdots x_N) \leqslant H(x_1)H(x_2)\cdots H(x_N)$. Show that the inequality can be strict.

Solution. **$\max(|xy|_v, 1) \leqslant \max(|x|_v, 1)\max(|y|_v, 1)$.** Let K be a number field containing $x_1, \ldots, x_N$. If $a, b \geqslant 0$, then $\max(ab, 1) \leqslant \max(a, 1)\max(b, 1)$, since the right side is at least ab and at least 1. By induction on N, for every $v \in \mathcal{P}_K$,

$$\max(|x_1 \cdots x_N|_v, 1) \leqslant \prod_{i=1}^{N} \max(|x_i|_v, 1).$$

Raise to the power n_v, multiply over v, and take the $[K : \mathbb{Q}]$-th root. For $x = 2$, the product $x \cdot x^{-1} = 1$ has height $1 < H(2)H(1/2) = 4$.

(2) Show that $H(x_1 + x_2 + \cdots + x_N) \leqslant N\, H(x_1)H(x_2)\cdots H(x_N)$, and that the factor N cannot be replaced by a smaller one.

Solution. **$\prod_v N^{\epsilon(v)n_v} = N^{[K:\mathbb{Q}]}$. Equality for $x_1 = \cdots = x_N = 1$.** Let K be a number field containing $x_1, \ldots, x_N$, and $v \in \mathcal{P}_K$. Let $\epsilon(v) := 1$ if v is archimedean and $\epsilon(v) := 0$ otherwise. The triangle inequality, or the ultrametric one, gives $|x_1 + \cdots + x_N|_v \leqslant N^{\epsilon(v)} \max_i |x_i|_v$. Since moreover $\max_i a_i \leqslant \prod_i \max(a_i, 1)$ for $a_i \geqslant 0$,

$$\max(|x_1 + \cdots + x_N|_v, 1) \leqslant N^{\epsilon(v)} \max\Big(\max_i |x_i|_v, 1\Big) \leqslant N^{\epsilon(v)} \prod_{i=1}^{N} \max(|x_i|_v, 1).$$

Raise to the power n_v and multiply over v. By part (1) of Proposition 1.1, for the extension $K/\mathbb{Q}$ and the archimedean place of $\mathbb{Q}$, $\sum_{v\in\mathcal{P}_K^\infty} n_v = [K:\mathbb{Q}]$. So the factors $N^{\epsilon(v)n_v}$ multiply to $N^{[K:\mathbb{Q}]}$, and the $[K:\mathbb{Q}]$-th root gives the claim. For $x_1 = \cdots = x_N = 1$, the left side is $H(N) = N$ and the right side is $N \cdot 1 = N$.

(3) Let $N, M \geqslant 0$, and let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ and $Q = [y_0 : \cdots : y_M] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. The Segre embedding sends (P, Q) to the point

$$S(P,Q) \coloneqq [x_0y_0 : x_0y_1 : \cdots : x_iy_j : \cdots : x_Ny_M] \in \mathbb{P}^{MN+M+N}(\overline{\mathbb{Q}}),$$

with the products x_iy_j in lexicographic order of (i,j). Show that $S(P,Q)$ does not depend on the choice of coordinates, and that $H(S(P,Q)) = H(P)H(Q)$.

Solution. $\mathbf{\max_{i,j}|x_iy_j|_v = |P|_v|Q|_v}$. There are $(N+1)(M+1) = MN+M+N+1$ products. If some $x_i \neq 0$ and some $y_j \neq 0$, then $x_iy_j \neq 0$. Rescaling the coordinates of P by λ and those of Q by μ rescales every product by $\lambda\mu$. Let K be a number field containing all x_i and y_j, and $v \in \mathcal{P}_K$. Since $|x_iy_j|_v = |x_i|_v|y_j|_v$, the maximum over (i,j) is attained at a pair where both factors are maximal. Raise to the power n_v, multiply over v, and take the $[K:\mathbb{Q}]$-th root.

(4) Let $N \geqslant 0$ and $d \geqslant 1$, let $M \coloneqq \binom{N+d}{N} - 1$, and let $f_0, \ldots, f_M$ be the distinct monomials of degree d in $X_0, \ldots, X_N$. The d-uple embedding sends $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ to $P^{(d)} \coloneqq [f_0(P) : \cdots : f_M(P)] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. Show that

$$H(P^{(d)}) = H(P)^d = H([x_0^d : \cdots : x_N^d]).$$

Solution. $\mathbf{\max_j|f_j(P)|_v = |P|_v^d = \max_i|x_i^d|_v}$. The monomials include $X_0^d, \ldots, X_N^d$, so $P^{(d)}$ has a nonzero coordinate, and rescaling P by λ rescales $P^{(d)}$ by λ^d. Let K be a number field containing $x_0, \ldots, x_N$, and $v \in \mathcal{P}_K$. Each $f_j(P)$ is a product of d coordinates of P, so $|f_j(P)|_v \leqslant |P|_v^d$, with equality for $f_j = X_i^d$ when $|x_i|_v = |P|_v$. The same holds for the coordinates x_i^d. No triangle inequality is involved, so no constant appears. Raise to the power n_v, multiply over v, and take the $[K:\mathbb{Q}]$-th root.

Remark. For $N = 1$ and $P = [x:1]$, the last equality is $H(x^d) = H(x)^d$, as in Problem 1.5(3). Compare [Sil09, Theorem VIII.5.6]: for a morphism F of degree d, only $H(F(P)) = H(P)^d$ up to bounded factors holds.

3. Heights and ideals

This section is [Sil09, Exercise 8.9], which extends [Sil09, Remark VIII.5.5] from $\mathbb{Q}$ to a number field. The description of the non-archimedean places is [HS00, Proposition B.1.3] and the discussion before it. For a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, let $\operatorname{ord}_\mathfrak{p}\colon K^\times \to \mathbb{Z}$ be the exponent of $\mathfrak{p}$ in the factorization of the fractional ideal $x\mathcal{O}_K$, and $\operatorname{Nm}\mathfrak{p} \coloneqq \#\mathcal{O}_K/\mathfrak{p}$. The norm extends multiplicatively to fractional ideals: $\operatorname{Nm}_{K/\mathbb{Q}}\mathfrak{a} \coloneqq \prod_\mathfrak{p}(\operatorname{Nm}\mathfrak{p})^{\operatorname{ord}_\mathfrak{p}\mathfrak{a}}$.

Proposition 3.1 (Non-archimedean places [HS00, Proposition B.1.3])**.** *The non-archimedean places of K correspond to the nonzero prime ideals of $\mathcal{O}_K$. If $v \in \mathcal{P}_K^0$ corresponds to $\mathfrak{p}$, then $|x|_v^{n_v} = (\operatorname{Nm}\mathfrak{p})^{-\operatorname{ord}_\mathfrak{p}(x)}$ for every $x \in K^\times$.*

Problem 3.2 (Heights and ideals)**.**

(1) Let $x_0, \dots, x_N \in K$, not all zero, let $P := [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$, and let $\mathfrak{b}$ be the fractional ideal of K generated by $x_0, \dots, x_N$. Show that

$$H_K(P) = \left(\mathrm{Nm}_{K/\mathbb{Q}}\,\mathfrak{b}\right)^{-1} \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}.$$

SOLUTION. **$\mathrm{ord}_\mathfrak{p}\,\mathfrak{b} = \min_i \mathrm{ord}_\mathfrak{p}(x_i)$, over the i with $x_i \neq 0$.** A sum of fractional ideals has, at each $\mathfrak{p}$, the least exponent of its summands, so $\mathrm{ord}_\mathfrak{p}\,\mathfrak{b} = \min_i \mathrm{ord}_\mathfrak{p}(x_i)$, over the i with $x_i \neq 0$. Let $v \in \mathcal{P}_K^0$ correspond to $\mathfrak{p}$. By Proposition 3.1, $|x_i|_v^{n_v} = (\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}(x_i)}$, which is largest when $\mathrm{ord}_\mathfrak{p}(x_i)$ is least. So $|P|_v^{n_v} = (\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}\,\mathfrak{b}}$. The exponent is zero for all but finitely many $\mathfrak{p}$, and the product over $\mathcal{P}_K^0$ is $\prod_\mathfrak{p}(\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}\,\mathfrak{b}} = (\mathrm{Nm}_{K/\mathbb{Q}}\,\mathfrak{b})^{-1}$.

(2) Let $K = \mathbb{Q}$. Deduce that if $x_0, \dots, x_N$ are coprime integers, then $H_\mathbb{Q}(P) = \max_i |x_i|$. More generally, show that if $\mathfrak{b}$ is principal, then P has coordinates in $\mathcal{O}_K$ that generate the unit ideal, and for such coordinates $H_K(P) = \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}$.

SOLUTION. **If $\mathfrak{b} = \lambda\mathcal{O}_K$, the coordinates x_i/λ generate $\mathcal{O}_K$.** If $\mathfrak{b} = \lambda\mathcal{O}_K$, then the elements x_i/λ generate $\lambda^{-1}\mathfrak{b} = \mathcal{O}_K$, so they lie in $\mathcal{O}_K$ and generate the unit ideal. For such coordinates the ideal of (1) is $\mathcal{O}_K$, of norm 1. Over $\mathbb{Q}$, every fractional ideal is principal, the ideal generated by coprime integers is $\mathbb{Z}$, and $\mathcal{P}_\mathbb{Q}^\infty$ is one place with $n_\infty = 1$.

(3) Let $K := \mathbb{Q}(\sqrt{-5})$, with $a := \sqrt{-5}$, and let $P := [2 : 1 + a] \in \mathbb{P}^1(K)$. `Magma` computes both sides of (1), and compares with its own logarithmic height of $(1 + a)/2$:

```
K<a> := QuadraticField(-5);
O := Integers(K);
xs := [K | 2, 1 + a];
b := ideal<O | xs>;
IsPrincipal(b), Norm(b);                                    // false 2
arch := &*[Max([Abs(Evaluate(x, v)) : x in xs])^LocalDegree(v)
           : v in InfinitePlaces(K)];
primes := &join[Support(x*O) : x in xs | x ne 0];
fin := &*[Norm(p)^(-Min([Valuation(x, p) : x in xs])) : p in primes];
RealField(10)!arch, fin;                                    // 6.000000000 1/2
[RealField(10) | AbsoluteLogarithmicHeight((1 + a)/2), Log(3)/2];
                                          // [ 0.5493061444, 0.5493061444 ]
```

Check the output by hand: show that $\mathfrak{b}$ has norm 2 and is not principal, and compute $H_K(P)$ and $H(P)$. Show that P has no coordinates in $\mathcal{O}_K$ that generate the unit ideal. Explain the last line of the output.

SOLUTION. **$H_K(P) = 6/2 = 3$, $H(P) = \sqrt{3}$, and $h((1 + a)/2) = (\log 3)/2$.** In $\mathcal{O}_K = \mathbb{Z}[a]$, $(1 + a)(1 - a) = 6$, and $\mathfrak{b}^2 = (4, 2 + 2a, -4 + 2a) = 2\,(2, 1 + a, -2 + a) = (2)$, since $(1 + a) - (-2 + a) = 3$ and $3 - 2 = 1$ lie in $(2, 1 + a, -2 + a)$. So $(\mathrm{Nm}\,\mathfrak{b})^2 = \mathrm{Nm}(2) = 4$, and $\mathrm{Nm}\,\mathfrak{b} = 2$. A generator of $\mathfrak{b}$ would be an element $u + wa$ of norm $u^2 + 5w^2 = 2$, and there is none. The field has one archimedean place, complex, with $n_v = 2$, where $|P|_v^2 = \max(|2|^2, |1 + a|^2) = \max(4, 6) = 6$. So $H_K(P) = 6/2 = 3$ and $H(P) = 3^{1/2}$.
Coordinates of P are $\lambda \cdot 2$ and $\lambda(1 + a)$ for some $\lambda \in K^\times$, and they generate $\lambda\mathfrak{b}$, which is not principal, hence not $\mathcal{O}_K$. In the output, `arch` $= 6$ is the archimedean product, and `fin` $= 1/2$ the non-archimedean one, which is $(\mathrm{Nm}\,\mathfrak{b})^{-1}$. Finally, $[(1 + a)/2 : 1] = [1 + a : 2]$, which is P with its coordinates swapped, so $h((1 + a)/2) = h(P) = (\log 3)/2$.

Remark. Choosing λ so that $\lambda\mathfrak{b}$ is one of finitely many representatives of the class group, (1) gives another proof of the second part of Theorem 1.4: the coordinates then lie in a fixed fractional ideal, with bounded absolute values at the archimedean places, and such elements are finitely many. Over $\mathbb{Q}$ the class group is trivial, which is why [Sil09, Example VIII.5.1] needs no product over the places.

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